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INVITED COMMENTARY

Mixing the Sheep and the Goats: Reframing the Interpretation of Acceleration Data in Sport

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Purpose: Acceleration is a continuous variable, and current practices of categorizing it into nonconsensual arbitrary thresholds may potentially lead to imprecise and biased data interpretation, as well as limiting interstudy comparisons. This commentary proposes a reframing of the interpretation of acceleration data and their use for athlete monitoring and exercise prescription. Although this applies to multiple sports, we will focus on the example of team sports in which these metrics are commonly used. ***Conclusion:*** Despite the recognized benefits of acceleration monitoring, 2 major limitations are evident. First, the common practice of classifying continuous acceleration data into discrete categories leads to a loss of relevant information, thus reducing the validity, reliability, and statistical power of analyses. Second, the use of arbitrary thresholds fails to account for individualized profiles, which are essential for monitoring. The limitations of these approaches are exacerbated by the lack of any objective justification for the thresholds used. We propose that acceleration be analyzed continuously, and, therefore, interpretation of data should be individualized. Accelerations should be analyzed as continuous variables rather than categorized into arbitrary thresholds. Individualized monitoring using players' maximum acceleration as a reference can provide a more accurate representation of players' physical demands.

Keywords: accelerometer, load monitoring, continuous variables, arbitrary thresholds, individualization

Acceleration data provide deep insights into the physical demands of various team sports. Previous studies have highlighted their importance, particularly in team sports, in which rapid changes in speed and direction are frequent.[1] Despite the benefits of monitoring acceleration, 2 relevant limitations warrant further discussion. Historically, advancements in tracking athletes' movements began with basic time–motion analysis, with tools like Global Positioning Systems (GPS) and accelerometers becoming accessible for more precise, real-time data collection. GPS reliance on satellite connections limits indoor usage and presents challenges for clubs with budget constraints, so accelerometers provide an alternative.[2] Within this context, the practice of categorizing continuous acceleration data, often into arbitrary thresholds (eg, accelerations above 2 m/s^2 or 4g), became standard to simplify analyses. However, categorizing continuous values results in the loss of information, which, in turn, lowers statistical power and reliability.[3] Today, this change from speed to acceleration thresholds highlights a need for continuous data analysis, moving beyond rigid classification systems.

Acceleration is a continuous variable but is often classified into discrete categories using nonconsensual arbitrary thresholds. This limitation is prevalent across various team sports, including American football,[4] situational sports,[5] basketball,[6] rugby,[7] and soccer.[8] For instance, accelerations were considered only above 4g,[5] whereas other authors considered movements requiring accelerations > 6g and > 8g[8] but without any rationale to support these selections. In this regard, there is growing recognition of the limitations of cut point-based analyses.[9] Although these thresholds are simple to apply and interpret, they hamper between-studies comparisons due to the wide variety of cut points used.[10] This challenge is exacerbated when authors exclude acceleration values above or below certain sensitive thresholds (eg, excluding accelerations lower than 1 m/s^2 because they are of very low intensity and very frequent), though the rationale behind these threshold selections is not always explicitly stated.[6] The same challenge has also been observed in speed data. For instance, different running velocity thresholds (ranging from 18 to 30 km/h) have been used to differentiate sprinting from high-speed running.[11] For a fast athlete, 18 km/h represents fast jogging, not sprinting. The challenges associated with acceleration data represent another manifestation of this same issue. In this context, we will use acceleration as an example.

These arbitrary thresholds also fail to account for individual differences,[12] which is essential for individualized monitoring. As the nature of the game involves frequent short accelerations,[13] neglecting individual variations in acceleration may lead to inaccurate estimations of match demands.[12] Therefore, this commentary stands that: (1) acceleration data are continuous and should be analyzed accordingly and (2) their interpretation should be individualized rather than relying on arbitrary thresholds.

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Analyzing Acceleration as a Continuous Variable

Acceleration has been often analyzed based on arbitrarily thresholds and without any justification for the zones created: >4*g*,[5] >3 m/s^2,[14] >4 m/s^2,[15] or 8 zones (<3*g*, >3*g* and <3.99*g*, >4*g* and <4.99*g*, >5*g* and <5.99*g*, >6*g* and <6.99*g*, >7*g* and <7.99*g*, >8*g* and <9.99*g*, and >10*g*).[16] Some researchers have also used previous studies of their authorship to justify the selected thresholds. However, neither of those studies has used a clear and acceptable justification for such thresholds because they were arbitrarily selected. Alternatively, a single study in rugby has proposed to set accelerations based on qualitative classifications ([1] dominant tackle/passive carry, [2] neutral tackle/carry, and [3] passive tackle/dominant carry).[7] Other rugby studies have considered acceleration categorizations based on 6 zones, with small variations in thresholds classification (very light, light to moderate impact, moderate to heavy impact, heavy impact, very heavy impact, and severe impact) highlighting the variation in approaches to classifying accelerations within the sport.[17,18] These arbitrary and inconsistent thresholds result in information loss by discretizing a continuous variable and limit between-studies comparability, as there is no universal agreement as to which thresholds should be used (see Figure 1).

Establishing criterion validity involves analyzing the data and comparing it with a known criterion value to allow for biologically meaningful interpretation of the data[19] independent of the frequency (in Hertz) or units (in meters per square second). However, research on this topic commonly adopts previously used thresholds without providing a clear rationale to support those decisions.[6,7] In contrast, a continuous analysis of acceleration would provide more accurate and easily comparable information.

Monitoring Acceleration Based on Individualized Reference Values

The individualization of training is crucial for optimizing players' development. In this sense, the selection of individualized monitoring methods represents a primary challenge for coaches and sport scientists. One possibility is to use the players' maximum acceleration efforts (see Figure 2).[20,21] The examination of continuous acceleration data provides reference values not only for maximum acceleration but also to understand the acceleration distribution curve, to analyze intravariability among players throughout the season, and to compare it with their usual range of accelerations.[22] However, caution is needed as the ability to estimate maximal acceleration through the acceleration–velocity profile remains questionable.[23]

This new approach implies regularly testing the players as maximum acceleration will likely change over the season,[12] as a consequence of the training process and more transient features, such as fatigue and contextual factors. Due to schedule limitations and the difficulty of incorporating testing into the teams' weekly plans, game and training data are used to update maximum acceleration, as is already done with maximum speed. This also allows new maximum reference values to be naturally reflected in the competitive context. Complementary tests, such as sprints, would be used sporadically for calibration purposes. The natural intraindividual

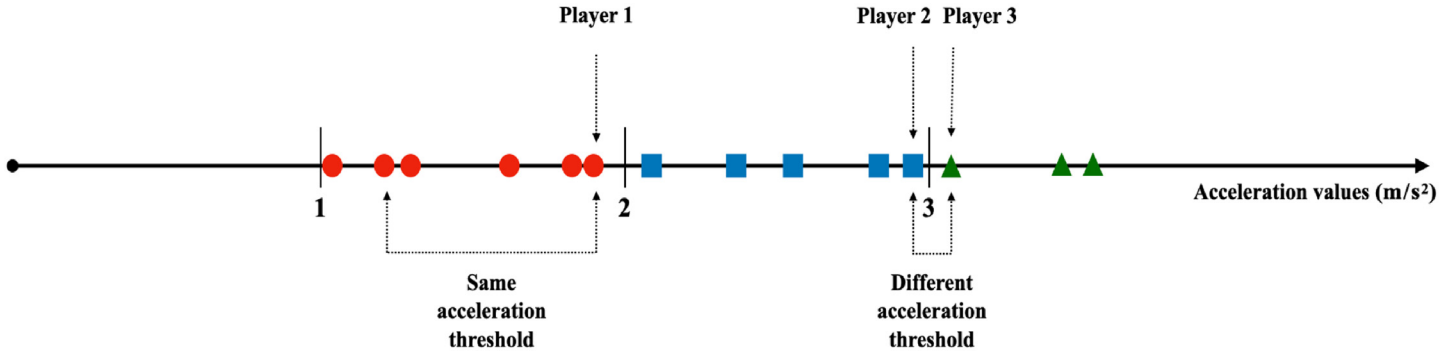

Figure 1 — Representation of the impact of acceleration arbitrary thresholds of 1 player wherein values that are significantly different are grouped together, whereas similar values are artificially separated.

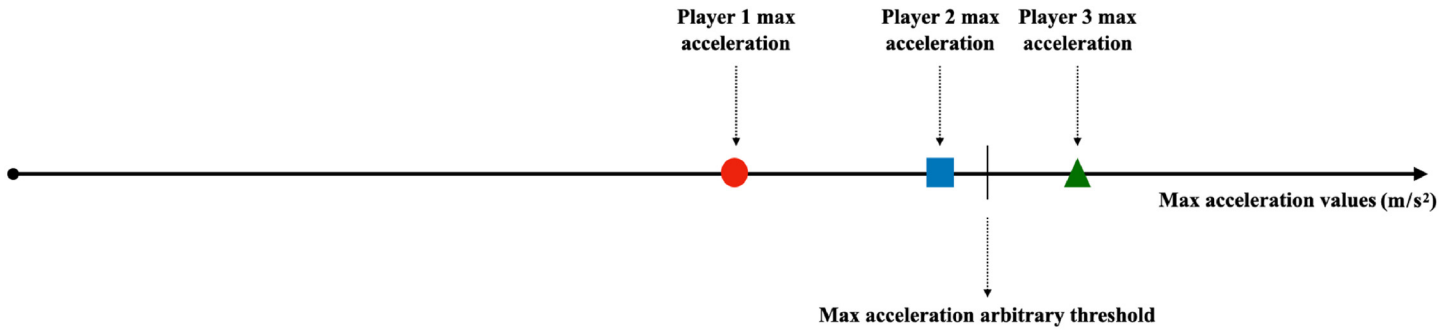

Figure 2 — Variation in the degree of proximity between each player's individual maximum acceleration and the arbitrary threshold.

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variability in acceleration values may be used to establish a range of values for maximum acceleration instead of a single-point value.

These new approaches offer valuable insights into individualized training load monitoring. However, some limitations may remain. First, heterogeneity in measurement methods with the use of accelerometers, Local Position Measurement or GPS-derived data,[12,20] would be a source of noise. Second, maximum acceleration capability may differ depending on the type of displacement, for example, linear versus curvilinear versus strong change of direction. Therefore, more refined and specific assessments related to these different abilities would likely be required as it is required with the current practices. Weather conditions (such as wind, temperature, and precipitation) as well as running surface, footwear, and clothing are additional factors that can influence acceleration analysis.[24]

Meanwhile, this individualized approach may improve understanding of individual players' demands rather than comparing them with other teams or players. The coaching staff would examine the distribution curve for the entire game, specific halves, and demands of certain drills, among other possibilities, to tailor the analyses according to specific contextual needs and goals.

A Potential Middle-Ground Solution? Individualized and Regularly Updated Thresholds

Threshold-based approaches result in information loss and may not reflect individual athletes' physical capacities.[12] Although speed thresholds are likely associated with metabolic thresholds (eg, an individual generally walks from 0 to 6 km/h),[25] metabolic thresholds for acceleration are not well established.[26] However, familiarity of the coaching staff with categorical methods of analysis plus the need for time-saving solutions may confer value to a middle-ground solution. Thus, the use of individualized thresholds, based on regularly updated individual maximum acceleration values, may provide a solution. Previously, high-intensity actions were classified as accelerations corresponding to ≥75% of the individual's maximum acceleration based on a Local Position Measurement system.[21] The idea was to individualize intensity thresholds tailored to a specific population or player and avoid the bias of underestimating actions with high initial speeds and overestimating those with low speeds. More recently, accelerations were calculated according to specific starting speed intervals (eg, <5, 5–10, and 10–15 km/h), and the effort intensities were established as very low (<25% of the maximal effort), low (25%–50%), moderate (50%–75%), and high (>75%).[27] Although the authors recommended practitioners to disregard the lowest intensity interval (<25%), this may imply the loss of potentially relevant information, which should be further verified.

This approach would take individual data into consideration, allowing practitioners and researchers to analyze performance based on the specific player, team, or group of players available. This method is not free of challenges and still categorizes a variable that is continuous but moves beyond the use of arbitrary thresholds with more cons than pros. This issue extends beyond acceleration to speed as using predefined starting speed intervals (eg, 0–5, 5–10, 10–15 km/h) is not optimal as the large differences within these intervals (eg, 5.1 vs 9.9 km/h) can significantly impact measured acceleration and speed performance.[28]

Practical Applications

We propose reframing the interpretation of acceleration analysis monitoring as a continuous variable and using its values for intraindividual, and not interindividual, comparisons (see Table 1). To individualize acceleration data, ongoing GPS and accelerometer data can be used to update values, potentially complemented with field-based tests. This may help to avoid underestimating or overestimating loads. Intravariability analysis throughout the season would allow players' training and match load monitoring and comparison. However, we still must distinguish acceleration efforts in different types of displacements, such as linear, curvilinear, and sudden changes of direction. In addition, the consideration of the speed at which acceleration starts is important, as it can significantly impact the final acceleration value. Our proposed approach may also be applicable to other metrics, such as speed.

Conclusion

This commentary proposes an approach to analysis of acceleration data that is continuous and individualized. Practitioners and researchers often use arbitrary and nonconsensual thresholds to understand players' acceleration demands, resulting in information loss and data that are not adjusted to each player. To account for the complexities of real-world practices, thresholds could potentially still be useful but calculated individually (in percentages) and updated regularly.

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Table 1 Reframing How Acceleration Data Are Analyzed in Sports

Reality	Problem	Solution
Acceleration classified into discrete categories	Loss of information, lower statistical power and reliability	Treating acceleration as a continuous variable
Arbitrary thresholds	No individualization and adjustment of values	Using acceleration based on individual values
Individual thresholds based on maximum acceleration	Maximum acceleration capability changes over the season and depending on the type of displacement (eg, linear vs curvilinear vs strong change of direction)	Individualized and regularly updated thresholds

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committee of the Faculty of Sport of the University of Porto approved the study (CEFADE 49 2022).

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